

1) Week 11: Numerical differentiation

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1.1) The distance x of a runner from a fixed point is measured in meters at intervals of half a second. The data obtained are:

t	x
0.0	0.0
0.5	3.65
1.0	6.80
1.5	9.90
2.0	12.15

Use forward, backward, and central differences to approximate the runner's velocity at times $t = 1.0$ s.

• Forward

$$x'_F(1.0) = \frac{x(1.5) - x(1.0)}{h} = \frac{9.9 - 6.8}{0.5} = 6.2$$

• Backward

$$x'_B(1.0) = \frac{x(1.0) - x(0.5)}{h} = \frac{6.8 - 3.65}{0.5} = 6.3$$

• Central

$$x'_C(1.0) = \frac{x(1.5) - x(0.5)}{2h} = \frac{9.9 - 3.65}{1.0} = 6.25$$

1.2) Suppose that you take samples of sea level heights at a rate of once every two hours, and you use the backward differences scheme to estimate the rate of change.

- a) Compute a bound on the size of the error incurred if sea level is sinusoidal with an amplitude of 3 feet and a period of 12 hours, meaning that it is generated by the function $f(t) = 3 \sin(\frac{2\pi}{12}t)$, with t measured in hours.
- b) How often should you take measurements if you wish the backward differencing error to not exceed 0.1 ft/hr?
- c) Repeat using the central differencing scheme.

$$a) |E_B(h)| \leq \frac{M_2}{2} h$$

$$\text{where } M_2 = \max_{t \in \mathbb{R}} |f''(t)|$$

$$f(t) = A \sin \omega t$$

$$f'(t) = A \omega \cos \omega t$$

$$f''(t) = -A \omega^2 \sin \omega t$$

$$\therefore M_2 = \max_{t \in \mathbb{R}} |-A \omega^2 \sin \omega t| = A \omega^2$$

$$\therefore |E_B(h)| \leq \frac{A \omega^2}{2} h = \frac{3 \left(\frac{2\pi}{12}\right)^2}{2} 2 = 0.822 \frac{\text{ft}}{\text{hr}}$$

$$b) \text{ Want } h \text{ such that } \frac{1}{2} A \omega^2 h < 0.1$$

$$\therefore \text{ need } h < \frac{2 \cdot 0.1}{A \omega^2} = \frac{2 \cdot 0.1}{3 \left(\frac{2\pi}{12}\right)^2} = 0.243 \text{ hr}$$

c) • Error bound with $h=2$.

$$f'''(t) = -A\omega^3 \cos \omega t \rightarrow M_3 = A\omega^3$$

$$|E_c| \leq \frac{M_3}{6} h^2 = \frac{A\omega^3}{6} h^2 = \frac{1}{6} 3 \left(\frac{2\pi}{12}\right)^3 \cdot 4 = 0.287 \frac{\text{ft}}{\text{hr}}$$

• h for error bound ≤ 0.1

$$\frac{1}{6} 3 \left(\frac{2\pi}{12}\right)^3 \cdot h^2 \leq 0.1$$

$$\therefore h \leq \sqrt{\frac{0.2}{(2\pi/12)^3}} = 1.18 \text{ hr}$$

2) Week 11: Numerical integration

2.1) Suppose we have taken the following samples of a function $f(x)$.

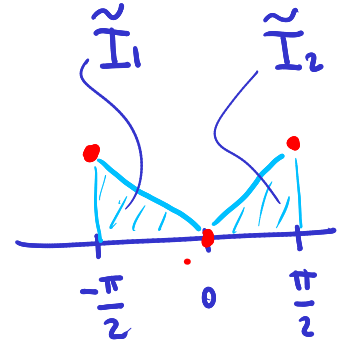
x	$f(x)$
$-\frac{\pi}{2}$	1
0	0
$\frac{\pi}{2}$	1

Use this data to estimate

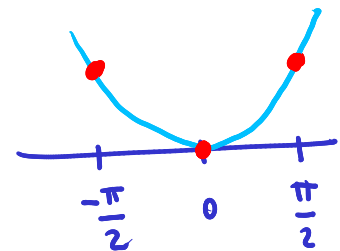
$$\int_{-\pi/2}^{\pi/2} f(x) dx$$

- a) with the composite trapezoidal rule
- b) with the composite Simpson's 1/3 rule

$$\begin{aligned} \text{a) } \tilde{I} &= \tilde{I}_1 + \tilde{I}_2 \\ &= \frac{h}{2} (f_1 + f_2) + \frac{h}{2} (f_2 + f_3) \\ &= \frac{\pi/2}{2} (1 + 0 + 0 + 1) = \frac{\pi}{2} \end{aligned}$$

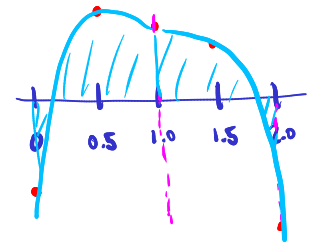


$$\begin{aligned} \text{b) } \tilde{I} &= \frac{h}{3} (f_0 + 4f_1 + f_2) \\ &= \frac{\pi}{6} (1 + 0 + 1) = \frac{\pi}{3} \end{aligned}$$



2.2) Use the following table of samples of $f(x)$ to approximate $\int_0^{2.5} f(x)dx$ using composite Simpson's 1/3 rule.

x	$f(x)$
0.0	-2.5
0.5	3.0
1.0	2.5
1.5	2.0
2.0	-4.5



$$\tilde{I} = \tilde{I}_1 + \tilde{I}_2$$

$$= \frac{h}{3} (f_0 + 4f_1 + f_2) + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

$$= \frac{0.5}{3} (-2.5 + 4 \cdot 3 + 2.5) + \frac{0.5}{3} (2.5 + 4 \cdot 2.0 - 4.5)$$

$$= 3$$

2.3) We wish to approximate the integral:

$$\int_0^3 x^4 dx$$

Find a bound on the approximation error of the following two techniques:

- a) Composite trapezoidal with $h = 0.2$.
- b) Composite Simpson's 1/3 rule with $h = 0.1$.

$$f(x) = x^4$$

$$f'(x) = 4x^3$$

$$f''(x) = 12x^2 \longrightarrow M_2 = 12 \cdot 9 = 108$$

$$a) \text{ error} \leq \frac{(b-a)h^2}{12} M_2 = \frac{3 \cdot 0.2^2}{12} \cdot 108 = 1.08$$

b)

$$f'''(x) = 24x$$

$$f''''(x) = 24 \longrightarrow M_4 = 24$$

$$\text{error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{3 \cdot (0.1)^4}{180} 24 = 4 \cdot 10^{-5}$$

2.4) We wish to approximate the integral of $f(x) = cx^2 + 3x + 4$ over the interval $[-6, 6]$ with a step size of $h = 0.5$. Find the maximum value of c such that the error is guaranteed not to exceed 1 when the approximation is made with

- a) the composite trapezoidal rule, or
- b) the composite Simpson's rule.

a) $f'(x) = 2cx + 3$

$$f''(x) = 2c \longrightarrow M_2 = 2|c|$$

$$\begin{aligned} \text{error} &\leq \frac{(b-a)h^2}{12} M_2 = \frac{(6+6)h^2}{12} \cdot 2|c| \\ &= 2h^2|c| \end{aligned}$$

Want this to be less than 1.

$$2h^2|c| < 1 \iff |c| < \frac{1}{2h^2} = \frac{1}{2 \cdot \frac{1}{4}} = 2$$

$\therefore$ maximum $C = 2$.

b) $f'''(x) = 0$

$$f^{(4)}(x) = 0 \longrightarrow M_4 = 0$$

$\therefore$ error $= 0$ for any value of c .

2.5) We wish to approximate the integral of $f(x) = e^{-x} \sin(x)$ on the interval $[0, 3]$ using a segment size of $h = 0.1$. Find an upper bound on the error made by

a) the composite trapezoidal rule

b) the composite Simpson's rule. (Note: Your estimate of M_4 does not have to be exact).

$$a) \quad f'(x) = -e^{-x} \sin x + e^{-x} \cos x = e^{-x} (\cos x - \sin x)$$

$$\begin{aligned} f''(x) &= -e^{-x} (\cos x - \sin x) + e^{-x} (-\sin x - \cos x) \\ &= -2e^{-x} \cos x \end{aligned}$$

$$M_2 = \max_{x \in [0, 3]} |f''(x)| = \max_{[0, 3]} 2e^{-x} \cos x = 2$$

$$\therefore \text{error} \leq \frac{(b-a)h^2}{12} M_2 = \frac{3(0.1)^2}{12} \cdot 2 = 0.005$$

$$b) \quad f'''(x) = 2e^{-x} \cos x + 2e^{-x} \sin x = 2e^{-x} (\sin x + \cos x)$$

$$\begin{aligned} f^{(4)}(x) &= -2e^{-x} (\sin x + \cos x) + 2e^{-x} (\cos x - \sin x) \\ &= -4e^{-x} \sin x \end{aligned}$$

$$M_4 = \max_{[0, 3]} 4e^{-x} |\sin x| \leq 4 \quad \dots \text{this is a conservative estimate, but it's the best I can do by hand.}$$

$$\therefore \text{error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{3(0.1)^4}{180} \cdot 4 = 6.6 \cdot 10^{-6}$$

2.6) For which of the following functions $f(x)$ is the estimate of

$$\int_0^{10} f(x) dx$$

using composite Simpson's rule with $h = 1$ guaranteed to produce an error less or equal to 0.5?

a) $f_1(t) = 3 \sin(t)$

b) $f_2(t) = e^{t/10}$

c) $f_3(t) = t^5$

d) $f_4(t) = 2t$

$$\text{In general, error} \leq \frac{(b-a)h^4}{180} M_4 = \frac{10 \cdot 1}{180} M_4 < 0.5$$

$$\text{Error} \leq 0.5 \text{ when } M_4 \leq \frac{180}{10} \cdot 0.5 = 9$$

a) $f_1' = 3 \cos t$

$$f_1'' = -3 \sin t$$

$$f_1''' = -3 \cos t$$

$$f_1^{(4)} = 3 \sin t \rightarrow M_4 = 3 \quad \checkmark$$

b) $f_2' = \frac{1}{10} e^{t/10}$

$$f_2'' = \frac{1}{100} e^{t/10}$$

$$f_2''' = \frac{1}{1000} e^{t/10}$$

$$f_2^{(4)} = \frac{1}{10000} e^{t/10} \rightarrow M_4 = \max_{(0,10)} \frac{e^{t/10}}{10000} = \frac{e}{10,000} < 9 \quad \checkmark$$

c) $f_3' = 5 t^4$

$$f_3'' = 20 t^3$$

$$f_3''' = 60 t^2$$

$$f_3^{(4)} = 120 t$$

$$\rightarrow M_4 = \max_{[0,10]} 120 t = 1200 > 9 \quad \times$$

d) $f_4^{(4)} = 0 \rightarrow M_4 = 0 \rightarrow \text{error} = 0 \quad \checkmark$

$$\begin{aligned}
 S_m = f(y_m) &= f((0.25, 1.25)) \\
 &= (1.25, (1-0.25)^2(1.25) - 0.25) \\
 &= (1.25, 0.453125) = \hat{S}
 \end{aligned}$$

$$\begin{aligned}
 \hat{y}[1] &= \hat{y}[0] + 0.5 \hat{S} \\
 &= (0, 1) + 0.5 (1.25, 0.453125) \\
 &= (0.625, 1.23)
 \end{aligned}$$